

## Database Technologies – Assignment 5

Use appropriate SQL functions to solve following queries. Note: To solve below queries, use “hr” database

1. Display all the first names in lower case and the last names in upper case as the single column separated with ' - '. Provide the meaningful name to the column.

```
SELECT CONCAT(LOWER(FIRST_NAME), ' - ', UPPER(LAST_NAME)) AS FIRST_Name  
FROM employees;
```

```
W2_99546_Pratiksha>SELECT CONCAT(LOWER(FIRST_NAME), ' - ', UPPER(LAST_NAME)) AS FIRST_Name  
-> FROM employees;  
+-----+  
| FIRST_Name |  
+-----+  
| ellen - ABEL |  
| sundar - ANDE |  
| mozhe - ATKINSON |  
| shelli - BAIDA |  
| amit - BANDA |  
| elizabeth - BATES |  
| sarah - BELL |  
| david - BERNSTEIN |  
| laura - BISSOT |  
| harrison - BLOOM |  
| hermann - BROWN |  
| alexis - BULL |  
| alexis - BULL |  
| matthew - WEISS |  
| jennifer - WHALEN |  
| david - WILLIAMS |  
| neena - YANG |  
| eleni - ZLOTKEY |  
+-----+  
107 rows in set (0.00 sec)
```

2. Display the first word of the job title. (Hint : Use substring\_index).

```
SELECT SUBSTRING_INDEX(JOB_TITLE, ' ',1)AS first_word_of_job FROM jobs;
```

```
W2_99546_Pratiksha>use hr_db;  
Database changed  
W2_99546_Pratiksha>SELECT SUBSTRING_INDEX(JOB_TITLE, ' ',1)AS first_word_of_job FROM jobs;  
+-----+  
| first_word_of_job |  
+-----+  
| Public |  
| Accounting |  
| Administration |  
| President |  
| Administration |  
| Accountant |  
| Finance |  
| Human |  
| Programmer |  
| Marketing |  
| Marketing |  
| Public |  
| Purchasing |  
| Purchasing |  
| Sales |  
| Sales |  
| Shipping |  
| Stock |  
| Stock |  
+-----+  
19 rows in set (0.01 sec)
```

3. Display the First name, last name and length of first name for the employees whose last name contains character 'b' anywhere after 3rd position.

```
SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME)
```

FROM employees

WHERE LAST\_NAME LIKE '---%B%';

```
W2_99546_Pratiksha>SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME) AS first_name_length
-> FROM employees
-> WHERE POSITION('b' IN LAST_NAME) > 3;
+-----+-----+-----+
| FIRST_NAME | LAST_NAME | first_name_length |
+-----+-----+-----+
| Gerald     | Cambrault | 6                 |
| Nanette    | Cambrault | 7                 |
| Nancy      | Gruenberg | 5                 |
| Susan      | Jacobs    | 5                 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

Select employee\_id,first\_name,last\_name from employees where first\_name LIKE '% %' OR last\_name LIKE '% %';

```
W2_99546_Pratiksha>Select employee_id,first_name,last_name from employees where first_name LIKE '% %' OR last_name LIKE '% %';
+-----+-----+-----+
| employee_id | first_name | last_name |
+-----+-----+-----+
| 112         | Jose Manuel | Urman     |
+-----+-----+-----+
1 row in set (0.00 sec)
```

5. Display first name, salary, and round the salary to thousands.

SELECT FIRST\_NAME,SALARY,ROUND(SALARY,-3)FROM EMPLOYEES AS ROUNDSALARY;

```
W2_99546_Pratiksha>SELECT FIRST_NAME,SALARY,ROUND(SALARY,-3)FROM EMPLOYEES AS ROUNDSALARY;
+-----+-----+-----+
| FIRST_NAME | SALARY   | ROUND(SALARY,-3) |
+-----+-----+-----+
| Steven     | 24000.00 | 24000             |
| Neena      | 17000.00 | 17000             |
| Lex        | 17000.00 | 17000             |
| Alexander  | 9000.00  | 9000              |
| Bruce      | 6000.00  | 6000              |
| David      | 4800.00  | 5000              |
| Valli      | 4800.00  | 5000              |
| Diana      | 4200.00  | 4000              |
| Nancy      | 12008.00 | 12000             |
| Daniel     | 9000.00  | 9000              |
| John       | 8200.00  | 8000              |
| Ismael     | 7700.00  | 8000              |
| Jose Manuel | 7800.00  | 8000              |
| Luis       | 6900.00  | 7000              |
| Den        | 11000.00 | 11000             |
| Alexander  | 3100.00  | 3000              |
| Shelli     | 2900.00  | 3000              |
| Sigal      | 2800.00  | 3000              |
| Jennifer   | 4400.00  | 4000              |
| Michael    | 13000.00 | 13000             |
| Pat        | 6000.00  | 6000              |
| Susan      | 6500.00  | 7000              |
| Hermann    | 10000.00 | 10000             |
| Shelley    | 12008.00 | 12000             |
| William    | 8300.00  | 8000              |
+-----+-----+-----+
107 rows in set (0.01 sec)
```

6. Display employee ID and the date on which he ended his previous job in the format as mentioned. (Format: Thursday, October 27th 2011)

Select employee\_id,end\_date,DATE\_FORMAT(end\_date, '%w,%m,%d,%y') AS JOB\_END\_DATE FROM job\_history;

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day. HINT: LAST\_DAY).

```
SELECT EMPLOYEE_ID,first_name,hire_date,date_add(last_day)(HIRE_date),interval +1
day)as first_salary from employees;
```

```
W2_99546_Pratiksha>SELECT EMPLOYEE_ID,first_name,hire_date,date_add(last_day(HIRE_date),interval +1 day)as first_salary from employees;
```

| EMPLOYEE_ID | first_name  | hire_date  | first_salary |
|-------------|-------------|------------|--------------|
| 100         | Steven      | 2013-06-17 | 2013-07-01   |
| 101         | Neena       | 2015-09-21 | 2015-10-01   |
| 102         | Lex         | 2011-01-13 | 2011-02-01   |
| 103         | Alexander   | 2016-01-03 | 2016-02-01   |
| 104         | Bruce       | 2017-05-21 | 2017-06-01   |
| 105         | David       | 2015-06-25 | 2015-07-01   |
| 106         | Valli       | 2016-02-05 | 2016-03-01   |
| 107         | Diana       | 2017-02-07 | 2017-03-01   |
| 108         | Nancy       | 2012-08-17 | 2012-09-01   |
| 109         | Daniel      | 2012-08-16 | 2012-09-01   |
| 110         | John        | 2015-09-28 | 2015-10-01   |
| 111         | Ismael      | 2015-09-30 | 2015-10-01   |
| 112         | Jose Manuel | 2016-03-07 | 2016-04-01   |
| 113         | Luis        | 2017-12-07 | 2018-01-01   |
| 114         | Den         | 2012-12-07 | 2013-01-01   |
| 115         | ...         | ...        | ...          |
| 200         | Jennifer    | 2013-09-17 | 2013-10-01   |
| 201         | Michael     | 2014-02-17 | 2014-03-01   |
| 202         | Pat         | 2015-08-17 | 2015-09-01   |
| 203         | Susan       | 2012-06-07 | 2012-07-01   |
| 204         | Hermann     | 2012-06-07 | 2012-07-01   |
| 205         | Shelley     | 2012-06-07 | 2012-07-01   |
| 206         | William     | 2012-06-07 | 2012-07-01   |

107 rows in set (0.02 sec)

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```
SELECT FIRST_NAME, HIRE_DATE, CONCAT(TIMESTAMPDIFF(YEAR, HIRE_DATE,
NOW()),'YEARS',TIMESTAMPDIFF(MONTH, HIRE_DATE, NOW())%12),
'MONTH',TIMESTAMPDIFF(DAY,HIRE_DATE,NOW())%31)AS EXPERINCED FROM EMPLOYEES;
```

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
SELECT EMPLOYEE_ID,FIRST_NAME,HIRE_DATE FROM EMPLOYEES WHERE
MONTH(HIRE_DATE)=8;
```

```
W2_99546_Pratiksha>SELECT EMPLOYEE_ID,FIRST_NAME,HIRE_DATE FROM EMPLOYEES WHERE MONTH(HIRE_DATE)=8;
```

| EMPLOYEE_ID | FIRST_NAME | HIRE_DATE  |
|-------------|------------|------------|
| 108         | Nancy      | 2012-08-17 |
| 109         | Daniel     | 2012-08-16 |
| 119         | Karen      | 2017-08-10 |
| 129         | Laura      | 2015-08-20 |
| 134         | Michael    | 2016-08-26 |
| 152         | Peter      | 2015-08-20 |
| 158         | Allan      | 2014-08-01 |
| 189         | Jennifer   | 2015-08-13 |
| 202         | Pat        | 2015-08-17 |

9 rows in set (0.00 sec)

10. Display the details of employees who joined before the year 2015.

```
SELECT EMPLOYEE_ID,HIRE_DATE FROM EMPLOYEES WHERE HIRE_DATE <' 2015-01-01';
```

```
W2_99546_Pratiksha>SELECT EMPLOYEE_ID,HIRE_DATE FROM EMPLOYEES WHERE HIRE_DATE <'2015-01-01';
```

| EMPLOYEE_ID | HIRE_DATE  |
|-------------|------------|
| 100         | 2013-06-17 |
| 102         | 2011-01-13 |
| 108         | 2012-08-17 |
| 109         | 2012-08-16 |
| 114         | 2012-12-07 |
| 115         | 2013-05-18 |
| 120         | 2014-07-18 |
| 122         | 2013-05-01 |
| 133         | 2014-06-14 |
| 137         | 2013-07-14 |
| 141         | 2013-10-17 |
| 145         | 2014-10-01 |
| 156         | 2014-01-30 |
| 157         | 2014-03-04 |
| 158         | 2014-08-01 |
| 174         | 2014-05-11 |
| 184         | 2014-01-27 |
| 192         | 2014-02-04 |
| 200         | 2013-09-17 |
| 201         | 2014-02-17 |
| 203         | 2012-06-07 |
| 204         | 2012-06-07 |
| 205         | 2012-06-07 |
| 206         | 2012-06-07 |

11. Display the number of days between system date and 1st January 2011.

```
SELECT TIMESTAMPDIFF(DAY, '2011-01-01', '2026-09-09')AS DIFF;
```

```
W2_99546_Pratiksha>SELECT TIMESTAMPDIFF(DAY, '2011-01-01', '2026-09-09')AS DIFF;
```

| DIFF |
|------|
| 5730 |

1 row in set (0.00 sec)

12. Display number of employees joined after 15th of any month.

```
SELECT COUNT(*)AS EMPLOYEE_COUNT FROM EMPLOYEES WHERE DAY (HIRE_DATE)>15;
```

```
W2_99546_Pratiksha>SELECT COUNT(*)AS EMPLOYEE_COUNT FROM EMPLOYEES WHERE DAY (HIRE_DATE)>15;
```

| EMPLOYEE_COUNT |
|----------------|
| 57             |

1 row in set (0.01 sec)

13. Display third highest salary of employees.

```
SELECT DISTINCT SALARY FROM EMPLOYEES ORDER BY SALARY DESC LIMIT 1 OFFSET 2;
```

```
W2_99546_Pratiksha>SELECT DISTINCT SALARY FROM EMPLOYEES ORDER BY SALARY DESC LIMIT 1 OFFSET 2;
```

| SALARY   |
|----------|
| 14000.00 |

1 row in set (0.00 sec)

## Assignment 5

Note: To solve below queries, use "spj" database

1. Display all the Suppliers, belonging to cities starting with the letter 'L'.

```
SELECT * FROM S WHERE CITY LIKE 'L%';
```

```
W2_99546_Pratiksha>SELECT * FROM S WHERE CITY LIKE 'L%';
+-----+-----+-----+-----+
| S# | Sname | status | city |
+-----+-----+-----+-----+
| S1 | Smith | 20 | London |
| S4 | Clark | 20 | London |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

2. Display all the Jobs, with the third letter in JNAME as 'n'.

```
SELECT * FROM J WHERE JNAME LIKE '__N%';
```

```
W2_99546_Pratiksha>SELECT * FROM K WHERE JNAME LIKE '__N%';
+-----+-----+-----+
| K# | Jname | CITY |
+-----+-----+-----+
| K2 | Punch | Rome |
| K4 | Console | Athens |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

3. Display all the Supplier names with the initial letter capital.

```
SELECT CONCAT(UPPER(LEFT(SNAME,1)),LOWER(SUBSTRING(SNAME,2)))AS NAMES
FROM S;
```

```
W2_99546_Pratiksha>SELECT CONCAT(UPPER(LEFT(SNAME,1)),LOWER(SUBSTRING(SNAME,2)))AS NAMES FROM S;
+-----+
| NAMES |
+-----+
| Smith |
| Jones |
| Blake |
| Clark |
| Adams |
+-----+
5 rows in set (0.00 sec)
```

4. Display all the Supplier names in upper case.

```
SELECT UPPER(SNAME)AS NAME FROM S;
```

```

ROM S at line 1
W2_99546_Pratiksha>SELECT UPPER(SNAME)AS NAME FROM S;
+-----+
| NAME |
+-----+
| SMITH |
| JONES |
| BLAKE |
| CLARK |
| ADAMS |
+-----+
5 rows in set (0.00 sec)

```

5. Display all the Supplier names in lower case.

```
SELECT LOWER(SNAME)AS NAME FROM S;
```

```

W2_99546_Pratiksha>SELECT LOWER(SNAME)AS NAME FROM S;
+-----+
| NAME |
+-----+
| smith |
| jones |
| blake |
| clark |
| adams |
+-----+
5 rows in set (0.00 sec)

```

6. Display the Supplier names and the lengths of the names.

```
SELECT SNAME,LENGTH(SNAME)AS LENGTH_OF_NAME FROM S;
```

```

W2_99546_Pratiksha>SELECT SNAME,LENGTH(SNAME)AS LENGTH_OF_NAME FROM S;
+-----+-----+
| SNAME | LENGTH_OF_NAME |
+-----+-----+
| Smith | 5 |
| Jones | 5 |
| Blake | 5 |
| Clark | 5 |
| Adams | 5 |
+-----+-----+
5 rows in set (0.00 sec)

```

7. Display the current day (e.g. Thursday).

```
SELECT DAYNAME (CURRENT_DATE());
```

```

W2_99546_Pratiksha>SELECT DAYNAME (CURRENT_DATE());
+-----+
| DAYNAME (CURRENT_DATE()) |
+-----+
| Thursday |
+-----+
1 row in set (0.02 sec)

```

8. Display the minimum Status in the Supplier table.

```
SELECT MIN(STATUS)AS MIN_STATUS FROM S;
```

```
W2_99546_Pratiksha>SELECT MIN(STATUS)AS MIN_STATUS FROM S;
+-----+
| MIN_STATUS |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)
```

9. Display the maximum Weight in the Parts table.

```
SELECT MAX(WEIGHT)AS MAX_WEIGHT FROM P;
```

```
W2_99546_Pratiksha>SELECT MAX(WEIGHT)AS MAX_WEIGHT FROM P;
+-----+
| MAX_WEIGHT |
+-----+
|          19 |
+-----+
1 row in set (0.00 sec)
```

10. Display the average Weight of the Parts.

```
SELECT AVG(WEIGHT)AS MIN_STATUS FROM P;
```

```
W2_99546_Pratiksha>SELECT AVG(WEIGHT)AS MIN_STATUS FROM P;
+-----+
| MIN_STATUS |
+-----+
|    15.1667 |
+-----+
1 row in set (0.00 sec)
```

11. Display the total Quantity sold for part 'P1'.

```
SELECT SUM(QTY)AS TOTAL_QUANTITY FROM SP WHERE `P#`='P1';
```

```
W2_99546_Pratiksha>SELECT SUM(QTY)AS TOTAL_QUANTITY FROM SP WHERE `P#`='P1';
+-----+
| TOTAL_QUANTITY |
+-----+
|          1000 |
+-----+
1 row in set (0.01 sec)
```

12. Display all the Supplier names (with 'la' replaced by 'ro'). HINT: REPLACE.

```
SELECT SNAME,REPLACE(SNAME, 'LA', 'RO')AS NEW_NAME FROM S;
```

```
W2_99546_Pratiksha>SELECT SNAME,REPLACE(SNAME, 'LA', 'RO')AS NEW_NAME FROM S;
+-----+-----+
| SNAME | NEW_NAME |
+-----+-----+
| Smith | Smith    |
| Jones | Jones    |
| Blake | Blake    |
| Clark | Clark    |
| Adams | Adams    |
+-----+-----+
5 rows in set (0.00 sec)
```